

Callable bonds

- 1) Theoretical issues
- 2) Valuation of callable bonds
- 3) **Measuring interest rate risk for callable bonds**

❖ Duration *effective Duration*

❖ Convexity

*Mac. Duration - assumes you get all CFs
assumes CFs aren't affected by i } inconsistent with CB
∴ we use effective duration*

Interest rate risk for Callable bonds

- The duration of a bond measures the sensitivity of the bond's full price to changes in the bond's yield to maturity or to changes in benchmark interest rates.
- For bonds with embedded options, the only appropriate duration measure is the curve duration measure known as **effective (or option-adjusted) duration**.



Effective duration indicates the sensitivity of the bond's price to a 100 bps (1%) parallel shift of the benchmark yield curve assuming no change in the bond's credit spread.

$$\text{Effective Duration} = \frac{(\overset{\downarrow}{PV_-}) - (\overset{\uparrow}{PV_+})}{2 \times (\Delta \text{Curve}) \times (\underset{\text{initial price}}{PV_0})}$$

Applicable to straight bond as well

Interest rate risk for callable bonds

Effective duration

Bond details; 3-year bond, Coupon rate of 4.25%; callable at par after one and two years from now: Calculate effective duration

- 1) Given a price (PV_0) of 101.000
- 2) We shift the par yield curve down by, say, 30 bps, generate a new interest rate tree, and then revalue the bond, PV_+ is 101.599. ^{Δ_{curve}}
- 3) We shift the par yield curve up by the same 30 bps, generate a new interest rate tree, and then revalue the bond, PV_- is 100.407. ^{PV_-}

$$\text{Effective duration} = \frac{101.599 - 100.407}{2 \times 0.003 \times 101.000} = 1.97$$

- ✓ (1) What does this duration mean? (2) How do you apply this duration
- ✓ Between a callable bond and a straight bond, which one is likely to have a higher duration.
- ✓ Is it possible for: Duration of a callable bond > Duration of vanilla bond
No. It's either = or lower

Duration of Callable vs Straight bond

- ❖ The effective duration of a callable bond cannot exceed that of the straight bond.
- ❖ *Either its lower or they are equal. Under what circumstances is it $<$ & $=$*



$$i > c_r$$

When interest rates are high relative to the bond's coupon rate, the callable and straight bonds have similar effective durations.

$$i < c_r$$

When interest rates fall, the effective duration of the callable bond is lower than that of the straight bond.



Interest rate risk for Callable bonds

Applying the effective duration estimate the price change.

❖ The percentage change in a bond's price, due to interest rates changes, is **proportional to its duration**:

$$\frac{\Delta P}{P} \approx -D^* \times \Delta y$$

$\frac{\Delta P}{P}$ is the % change in the bond's price, Δy is the change in YTM.

Interest rate risk for Callable bonds

Shortcomings of duration

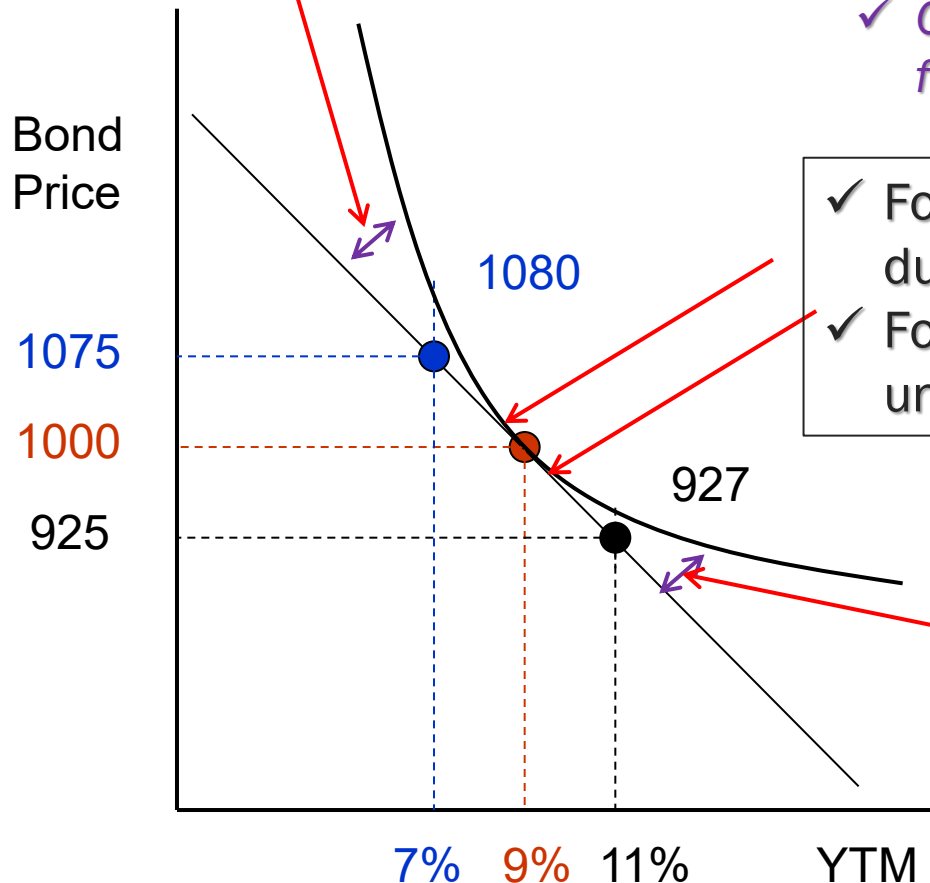
- ❖ Effective duration suffers from the same shortcomings of Macaulay duration.
- ❖ It is an inaccurate measure of bond price sensitivity
 - ✓ It assumes a linear relationship between bond prices and yields
 - ✓ Yet the bond price-yield relationship is not linear – its convex (convexity)
- ❖ It assumes a parallel shift in the yield curve

∴ you need convexity

Interest rate risk

Analysis of the duration measure

Error not measured by duration



For me

- ✓ Shortcomings – inaccurate, underestimates, accurate for small changes not large
- ✓ Why duration fails
- ✓ Conclusion and way forward

- ✓ For small changes the duration is accurate
- ✓ For large changes it underestimates.

Error not measured by duration

Convexity for Callable bonds

Convexity of callable bonds

- ❖ Relationship between bond price and yield is not linear.
- ❖ Convexity is the curvature of the price/yield relationship

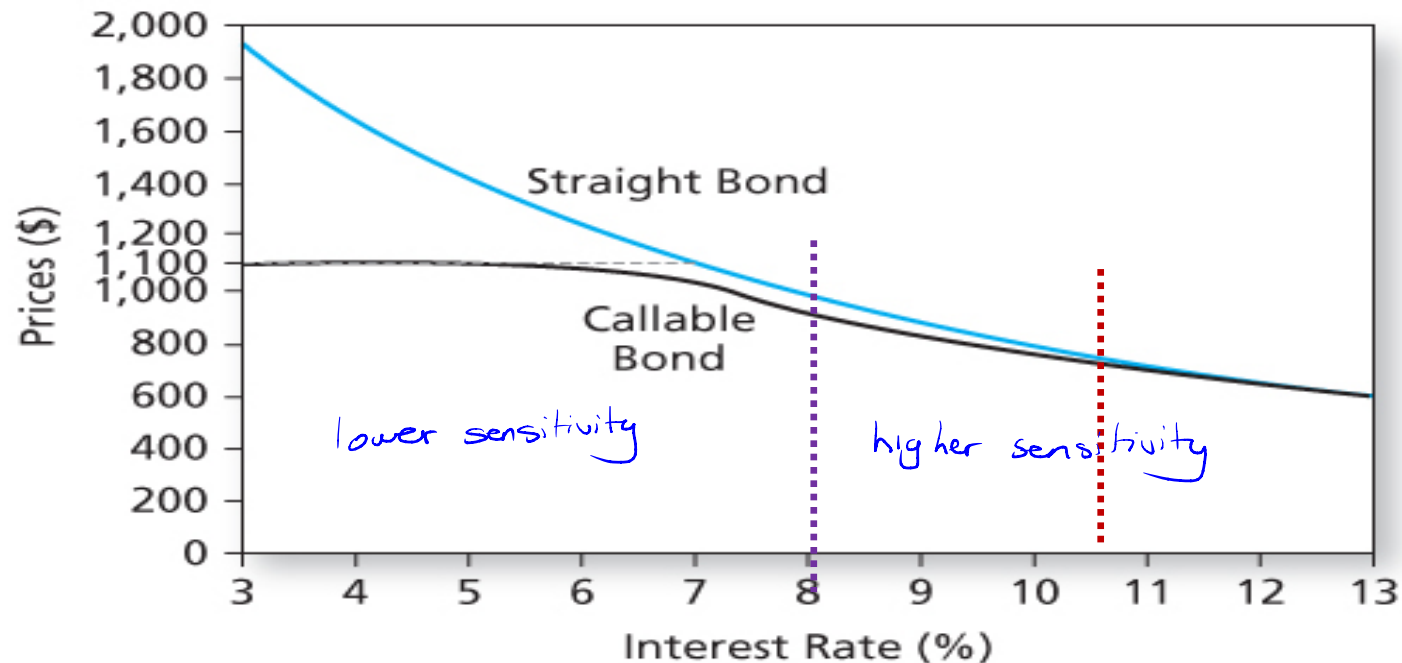


Figure 14.4 Bond prices: Callable and straight debt. Coupon = 8%; maturity = 30 years; semiannual payments.

Convexity for Callable vs Straight bonds

- 
- When interest rates are high, the callable and straight bond experience very similar positive convexity.

- 
- The effective convexity of the callable bond turns negative when the call option is near the money.

Convexity of callable bonds

Positive convexity effect in general

- ❖ Size of the increase in bond price resulting from a given decrease in yield is **greater** than the drop in price resulting from a similar increase in yield.

Coupon rate	Face value	YTM	Price	Price change
5%	R1,000	3%	R1,170,60	R170,60
5%	R1,000	5%	R1,000	
5%	R1,000	7%	R859,53	(140,47)

- ❖ *High positive convexity implies high interest rate risk*
- ❖ *Callable bonds can have both positive and negative convexity*

Negative convexity for callable bonds

Callable bonds exhibiting negative convexity at lower interest rates relative to the coupon rate

❖ YTM=5%, CR=5%, FV=R1000, Price =R1,000, call price is R1,050.

Coupon rate	Face value	YTM	Price	Price change
5%	R1,000	3%	R1,170,60 Called @ R1,050	R50
5%	R1,000	5%	R1,000	
5%	R1,000	7%	R859,53	(140,47)

Negative convexity: An equal adjustment in yield (up & down) results in a greater price depreciation than the price appreciation.

Positive convexity for callable bonds

Callable bonds exhibiting positive convexity at higher interest rates relative to the coupon rate

❖ YTM=5%, CR=5%, FV=R1000, Price =R1,000, call price is R1,050.

↪ high compared to Cr

Coupon rate	Face value	YTM	Price	Price change
5%	R1,000	28%	R248,15	R21,03
5%	R1,000	30%	R227,12	
5%	R1,000	32%	R208,79	(18,33)

Effective convexity for callable bonds:

Effective convexity is calculated for callable, similar to the straight bond.

$$EffConv = \frac{(PV_-) + (PV_+) - 2 \times (PV_0)}{(\Delta curve)^2 \times PV_0} \quad \text{given}$$

You are given 3-year 4.25% bond callable at par in one year and two years from now. Assume the bond's current full price is 100.785. Given 30 bps shifts in the benchmark yield curve, the resulting PV_- and PV_+ are 101.381 and 100.146, respectively, Calculate the effective convexity.

$$EffConv = \frac{101.381 + 100.146 - 2 \times 100.785}{(0.003)^2 \times 100.785} = -47.41$$

Q: Under what circumstances will a vanilla bond and callable bond have similar convexity

2:27:00 listen here for tips

10 marks
RCY from
last lecture

Interest rate risk for callable bonds

News article $\Rightarrow$ what shift (how event results in the shift)
 $\hookrightarrow$ will be posted on Amathuba what strategy would you implement

- 1) How do you measure interest rate risk for callable bonds.
- 2) Compare and contrast interest rate risk for callable bonds vs Vanilla bonds:
- 3) High convexity is an attractive feature for bond traders

Blum CB, PB ^{highest} & VB, which highest & lowest conv. & justify
 $\hookrightarrow$ lowest